🧪 8TH GRADE SCIENCE: CHEMISTRY REVIEW

🌍 Classifying Matter `S8P1a`

Matter = anything that has **mass** and takes up **space (volume)**.

PURE SUBSTANCES

Only ONE kind of particle. Always the same.

- **Element:** 1 type of atom (Au, O, Fe)
- **Compound:** 2+ elements chemically joined (H_2O, CO_2, NaCl)

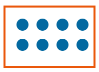
MIXTURES

2+ substances physically combined — keep their own properties.

- **Homogeneous:** looks uniform (saltwater, air)
- **Heterogeneous:** you can see parts (pizza, salad)

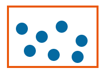
🔥 ❄️ Phases of Matter & Thermal Energy `S8P1b`

Adding heat makes particles move faster → phase changes! **Removing heat** slows them down.



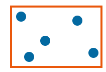
SOLID

packed tight, vibrate in place, fixed shape & volume



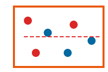
LIQUID

close but slide past each other, takes container shape



GAS

far apart, move fast in all directions, fills container



PLASMA

super-hot, charged particles — stars, lightning, neon signs

← **melt** → **evaporate** → **ionize** (add heat) ← **freeze** ← **condense** ← **de-ionize** (remove heat)

⚖️ Physical vs. Chemical — Properties & Changes `S8P1c & 1d`

PHYSICAL (no new substance)

Properties: color, shape, size, density, state, texture, melting/boiling point, magnetism, solubility

Changes: cutting, melting, freezing, boiling, dissolving, crushing, tearing

CHEMICAL (NEW substance forms!)

Properties: flammability, reactivity with water/acid, ability to rust or tarnish

Signs of change: color change • gas/bubbles • odor • light or heat • precipitate (solid forms)

✏️ ACTIVITY 1: Physical or Chemical?

Label each change as **P** (physical) or **C** (chemical):

____ 1. Ice melting in a glass
____ 2. Wood burning in a fireplace
____ 3. Iron rusting on a car
____ 4. Cutting paper with scissors
____ 5. Baking a cake
____ 6. Salt dissolving in water
____ 7. Milk turning sour
____ 8. Tearing a napkin
____ 9. Fireworks exploding

🧲 The Periodic Table `S8P1e`

The periodic table is organized by **atomic number**. Rows are **periods**, columns are **groups/families**.

H								He
Li	Be		B	C	N	O	F	Ne
Na	Mg		Al	Si	P	S	Cl	Ar
K	Ca	Transition Metals	Ga	Ge	As	Se	Br	Kr

☐ Metals ☐ Metalloids ☐ Nonmetals ☐ Noble Gases ☐ Alkali Metals ☐ Halogens

METALS (left + middle) — shiny, conduct heat & electricity, bendable, usually solid

METALLOIDS (stair-step) — in-between properties; used in electronics (Si)

NONMETALS (right side) — dull, poor conductors, brittle, often gas/liquid

🔍 Reading a Periodic Table Square `S8P1e`

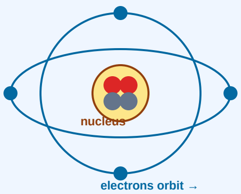
6
C
Carbon
12.01

On the Square	What It Is	How to Use It
6 (top)	**Atomic Number**	# of **protons** (also # of electrons in a neutral atom)
C (middle)	Element Symbol	1–2 letter code; 1st letter capital, 2nd lowercase
Carbon	Element Name	Full name of the element
12.01 (bottom)	**Atomic Mass**	Round to nearest whole # = **Mass Number**

⭐ **Counting Particles:** **Protons = Atomic #** • Electrons (neutral atom) = Atomic # • **Neutrons = Mass # − Atomic #**

🧪 *Flip over for atoms, counting atoms, chemical reactions & balanced equations!* →

⚛️ Inside the Atom S8P1e



Particle	Charge	Location	Mass
Proton	+ (positive)	Nucleus	1 amu
Neutron	0 (neutral)	Nucleus	1 amu
Electron	− (negative)	Orbits nucleus	tiny

Example — Oxygen: atomic # = 8, mass # = 16 → **8 protons**, 8 electrons, 16 − 8 = **8 neutrons**

✏️ ACTIVITY 2: Periodic Table Detective

Use the periodic table to fill in the blanks:

Element	Symbol	Atomic #	Mass #	Protons	Neutrons	Electrons	Metal / Metalloid / Nonmetal?
Sodium	Na	11	23	____	____	____	________________
Fluorine	F	9	19	____	____	____	________________
Silicon	Si	14	28	____	____	____	________________
Aluminum	Al	13	27	____	____	____	________________

🧬 Molecules & Chemical Formulas S8P1e

A **molecule** is 2 or more atoms bonded together. A **chemical formula** uses symbols + numbers to show which atoms and how many.

🔢 Reading Subscripts & Coefficients

2 H₂ O → **Total H: 2×2 = 4**
Total O: 2×1 = 2

Coefficient = # of molecules
Subscript = # atoms

To count atoms: multiply the coefficient by each subscript. If no number is written, it's **1**.

Examples of Counting Atoms

Formula	Atoms of each element
CO_2	1 C, 2 O (3 total)
H_2SO_4	2 H, 1 S, 4 O (7 total)
3 NaCl	3 Na, 3 Cl
2 $C_6H_{12}O_6$	12 C, 24 H, 12 O

⚖️ Conservation of Matter & Balanced Equations S8P1f

Law of Conservation of Matter: In a chemical reaction, matter is NOT created or destroyed. The **atoms you start with = the atoms you end with** (just rearranged!).

Parts of a Chemical Equation

$$2\ H_2 + O_2 \rightarrow 2\ H_2O$$

REACTANTS (before arrow) → **PRODUCTS** (after arrow)
The arrow means "reacts to form" or "yields."

✅ Is It Balanced? Count Both Sides!

Equation	Left side	Right side	Balanced?
$2H_2 + O_2 \rightarrow 2H_2O$	4 H, 2 O	4 H, 2 O	✓ YES
$H_2 + O_2 \rightarrow H_2O$	2 H, 2 O	2 H, 1 O	✗ NO
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	1 C, 4 H, 4 O	1 C, 4 H, 4 O	✓ YES

🔑 **KEY:** To balance, change **coefficients** (big numbers in front) — NEVER change subscripts! Changing a subscript changes what the molecule IS.

✏️ ACTIVITY 3: Count the Atoms

Write how many atoms of each element:

1. H_2O → H = ___ O = ___
2. $C_6H_{12}O_6$ → C = ___ H = ___ O = ___
3. 3 NH_3 → N = ___ H = ___
4. 2 $Ca(NO_3)_2$ → Ca = ___ N = ___ O = ___
5. Al_2O_3 → Al = ___ O = ___

✏️ ACTIVITY 4: Balanced or NOT?

Circle "Yes" or "No" for each:

1. $N_2 + 3H_2 \rightarrow 2NH_3$ Y / N
2. $Mg + O_2 \rightarrow MgO$ Y / N
3. $2Na + Cl_2 \rightarrow 2NaCl$ Y / N
4. $H_2 + Cl_2 \rightarrow HCl$ Y / N
5. $2H_2O_2 \rightarrow 2H_2O + O_2$ Y / N

📝 I Can… (Self-Check)

- ☐ Tell pure substances from mixtures
- ☐ Describe how particles move in each state of matter
- ☐ Tell physical changes from chemical changes
- ☐ Identify metals, metalloids, and nonmetals
- ☐ Read a periodic table square (symbol, atomic #, mass #)
- ☐ Find # of protons, neutrons, and electrons
- ☐ Count atoms in a chemical formula
- ☐ Decide if a chemical equation is balanced